Wildchrokie ms

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Introduction

Methods

Common garden setup

We collected seeds from four provenances across a 3.5° latitudinal gradient (Figure 1 and Table 1) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings. In the spring of 2017, we planted them outside in the garden located at the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020. We initially collected seeds of 18 species of northeastern North American trees and shrubs. We exclude the shrubs, and the five species with low survivorship in the analyses. Thus, we have four species in the *Betulaceae* family (Table 1).

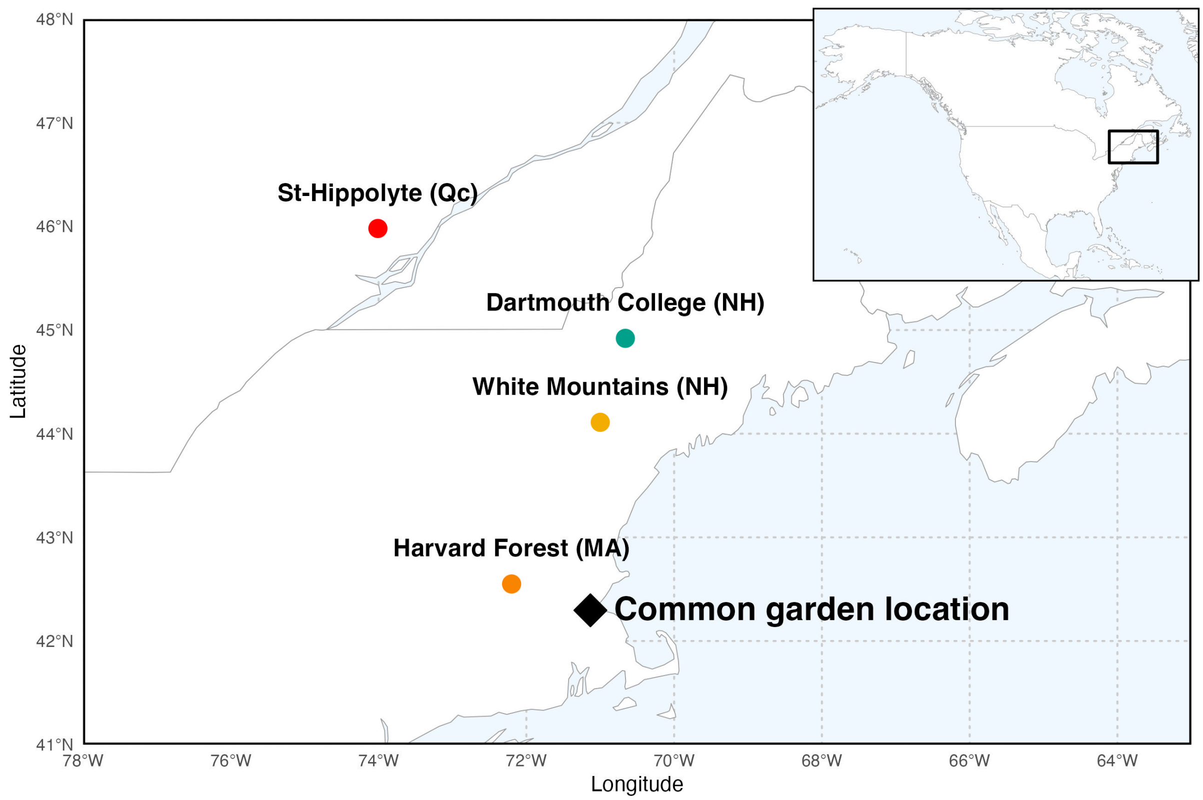

Figure 1: Provenance are depecticted as circles, while the star denotes the location of the common garden the seeds accross a latitudinal gradient. The common garden location is where the study took place at the Weld Research Building (Boston, MA).

Phenological monitoring and sample collection

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December (Figure S1). See (Buonuito, unpublished) for more details about the phenological monitoring. In the spring of 2023, we collected cross-sections for 78 individual trees, cutting the trunk just above the root collar of most trees. We also collected tree cores for 43 individuals using standard dendrochonological methods and stored them in modified plastic straws to allow aeration. Both the cross-sections and cores were left to dry at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted the cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits: 150, 300, 400, 600, 800, 1000. We scanned the cores and cross-sections at a resolution of 6250 dpi, with a high-resolution treering scanner (Fong, unpublished), we then used the digitized images to measure the tree ring widths with ImageJ (Schneider *et al.*, 2012). For individuals for which we had cross-sections, we measured ring widths at three standard locations on the cross-section, at 120° from each other. We averaged the three measurements, to yield one ring width per year, per individual. For the 8 individuals for which we only had cores, we measured only one ring width for that sample. When we had cross-sections and cores for an invidividual, we discarded the measurement from the core.

We performed visual cross dating, but did not perform statistical crossdating because of the short chronologies that limit the capacity of these analyses (Raden *et al.*, 2020). Given the short lifespan of the trees in our dataset, we did not detrend the ring widths. However, we included a year intercept that accounts for other factors that could affect growth, such as age allometry.

Data analyses

To understand how growth shifted with season lengths we used linear Bayesian hierarchical models. With these models, we estimated ring width as a function of four different season predictors. First, the thermal season represents the number of growing degree days (GDD) calculated as the mean daily temperature above 5°C accumulated between leafout and budset, calculated for each tree per year. Second, the calendar season or growing season length (GSL) is the budset date subtracted by the leafout date. Finally, to better understand trees responses to the calendar season, we used leafout as the start of the season (SOS) and budset for the end of season (EOS) as separate predictors that represent the boundaries of the growing season.

$$
\begin{aligned}
ln(ringwidth)_i &\sim N(\mu, \sigma_y) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{site}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha &\sim N(1, 3) \\
\alpha_{\text{species}} &\sim N(0, 6) \\
\alpha_{\text{site}} &\sim N(0, \sigma_{\alpha\,\text{site}}) \\
\alpha_{\text{treeid}} &\sim N(0, \sigma_{\alpha\,\text{treeid}}) \\
\beta_{\text{species}} &\sim N(0, 0.5) \\
\sigma_{\alpha\,\text{site}} &\sim N(0, 1) \\
\sigma_{\alpha\,\text{treeid}} &\sim N(0, 1) \\
\sigma_y &\sim N(0, 3)
\end{aligned}
$$

where i is each observation of the dataset. Intercepts on species ($\alpha_{species}$) and year (α_{year}) and slope on species ($\beta_{species}$) were not pooled (fixed effects), and we partially pooled (random effects) site (α_{site}) and treeid (α_{treeid}) to account for the similarity of the measurements within these groups. We ran four chains with 2000 warmup iterations each, which we discarded, and 2000 sampling iterations, which we kept for posterior distribution estimates. The models did not have any divergent transitions, had a $\hat{R}$ below 1.01, and values of effective sample size (n_{eff}) greater than 10% of the model iterations. We fit ln(ring width) so the data distribution was normally distributed. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

To account for differences in temperature across years, we used the daily climate data from the climate station located at Weldhill research station for 2018 and 2019 (`https://arboretum.harvard.edu/research/data-resources/weather/`). Because the data was not available for that station after August 2020, we used the daily climate data from the weather station "Boston (Weld Hill)" of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (`https://newa.cornell.edu/`

all-weather-data-query/). We calculated daily growing degree days (GDD) by subtracting a 5°C threshold from the mean daily temperature.

As a supplementary analyses, we ran the models using basal area increment (BAI), as the response variable. For this, we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring:

$$\pi \times (\text{Inner radius}^2 - \text{Outer radius}^2)$$

We then averaged across the three BAI for each ring and fit the models with the four different predictors.

Table 1: Provenance coordinates and count per species and provenances. We had a total of 81 individual trees

Provenance	Longitude	Latitude	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	-72.2	42.5	7	0	4	7
White Mountains (NH)	-71.0	44.1	6	7	0	9
Dartmouth College (NH)	-70.7	44.9	8	7	2	9
St-Hippolyte (Qc)	-74.0	46.0	8	6	1	0

Results

Summary basic information

Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals. At the common garden location, the mean summer temperature was similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C). There was no drought for those three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September. The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018). The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018). Finally, annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S6).

Does tree growth increase with longer growing seasons?

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. Indeed, warmer thermal season increased growth for *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.18, 0.7]), *B. populifolia* ($\beta = 0.22$, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.24]), but *A. incana* trended in the same direction ($\beta = 0.07$, UI_{90} [-0.04, 0.17]) (Figure 2a and e) (TableS1).

Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0, 0.1]). However, of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI_{90} [0.05, 0.32]), but *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.02, 0.09]), and *B. populifolia* ($\beta = 0.05$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 2b and f) (Table S1).

On a scaled axis, both predictors were similarly predictive (FigureS8). Then, the results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S3).

Did a later start and end of season increase growth?

A longer calendar season increased growth for two species, but only one also grew more with an earlier start of season (*A. incana*: $\beta =$ -0.07, UI_{90} [-0.14, -0.01]) (Figure 2c and g) (Table S1). Conversely, *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction ($\beta = 0.1$, UI_{90} [0, 0.2]) (Figure 2c and g) (Table S1). In addition, three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, UI_{90} [0.02, 0.17]), with *B. papyrifera* ($\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 2d and h) (Table S1).

The estimates were unchanged when using the full (SOS $n = 239$ and $n = 230$) or restricted datasets ($n = 227$) (Figure S2a).

How did tree growth change across years, provenances and tree ids?

We did not find any clear effect of year on growth (Figure S6) and there is no apparent differences of growth looking across species (Figure S3 and S4). Likewise, growth did not change across any provenances (Figure 3) (Table S2), and the variation across sites ($\sigma = 0.18$, UI_{90} [0.02, 0.5) and tree ids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]) was similar.

Discussion

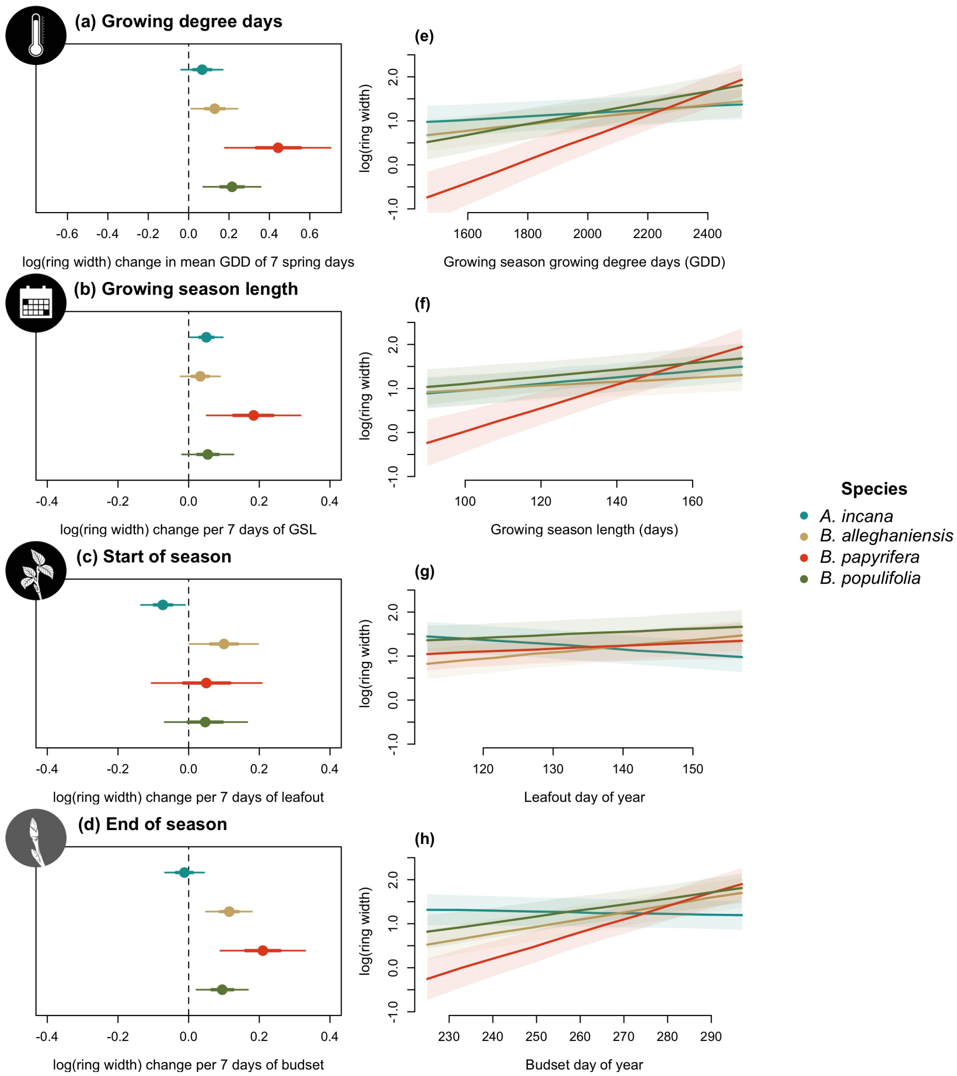

Figure 2: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (174.05) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

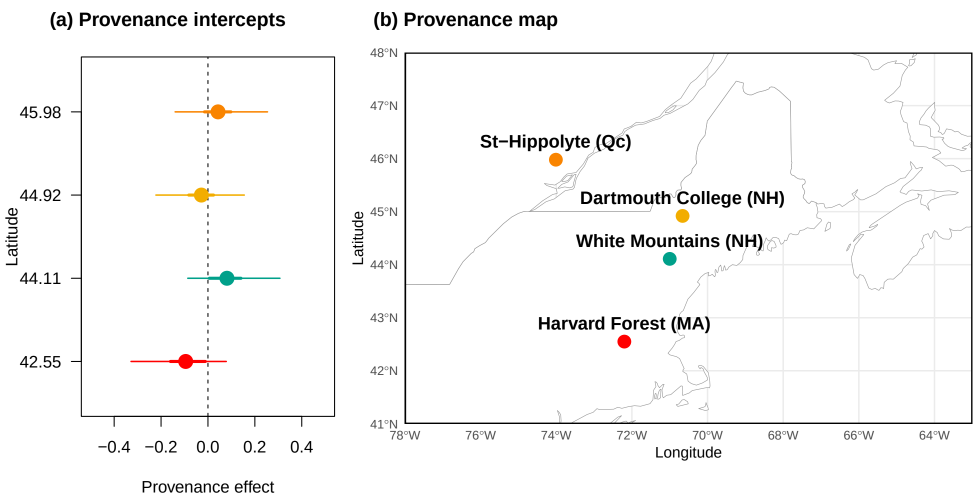

Figure 3: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

References

Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P. & Riddell, A. (2017). *Stan* : A Probabilistic Programming Language. *Journal of Statistical Software*, 76.

Raden, M., Mattheis, A., Spiecker, H., Backofen, R. & Kahle, H.P. (2020). The potential of intra-annual density information for crossdating of short tree-ring series. *Dendrochronologia*, 60, 125679.

Schneider, C.A., Rasband, W.S. & Eliceiri, K.W. (2012). NIH Image to ImageJ: 25 years of image analysis. *Nature Methods*, 9, 671–675.

Stan Development Team (2025). RStan: the R interface to Stan.

Supplemental

Supplemental Methods

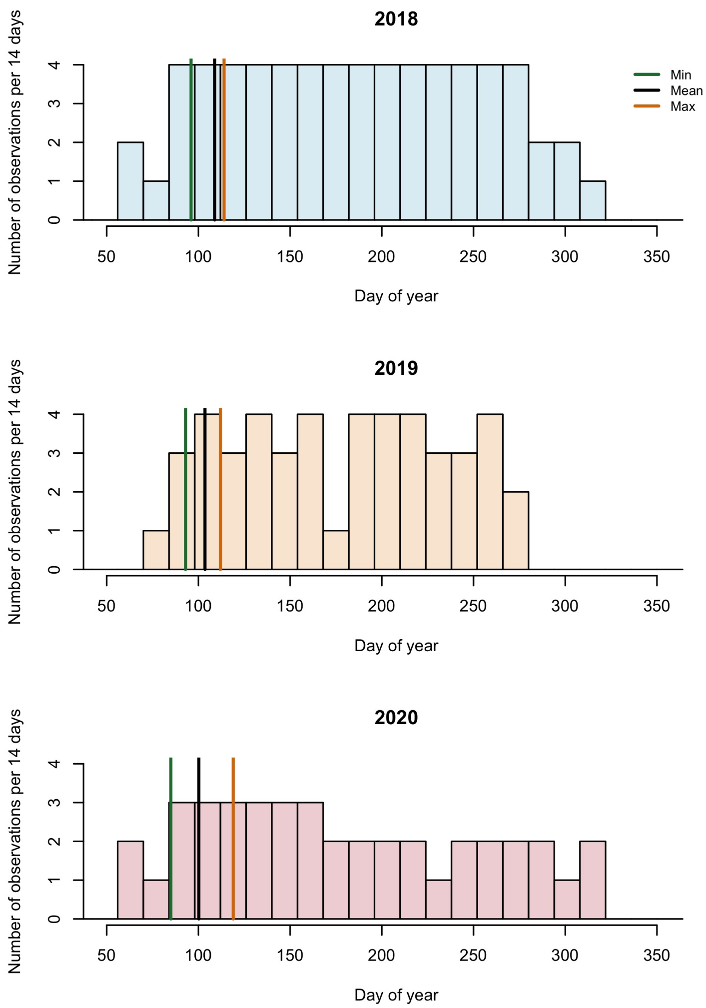

Figure S1: Phenology observatons bi-weekly.

Supplemental Results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]	-74.03

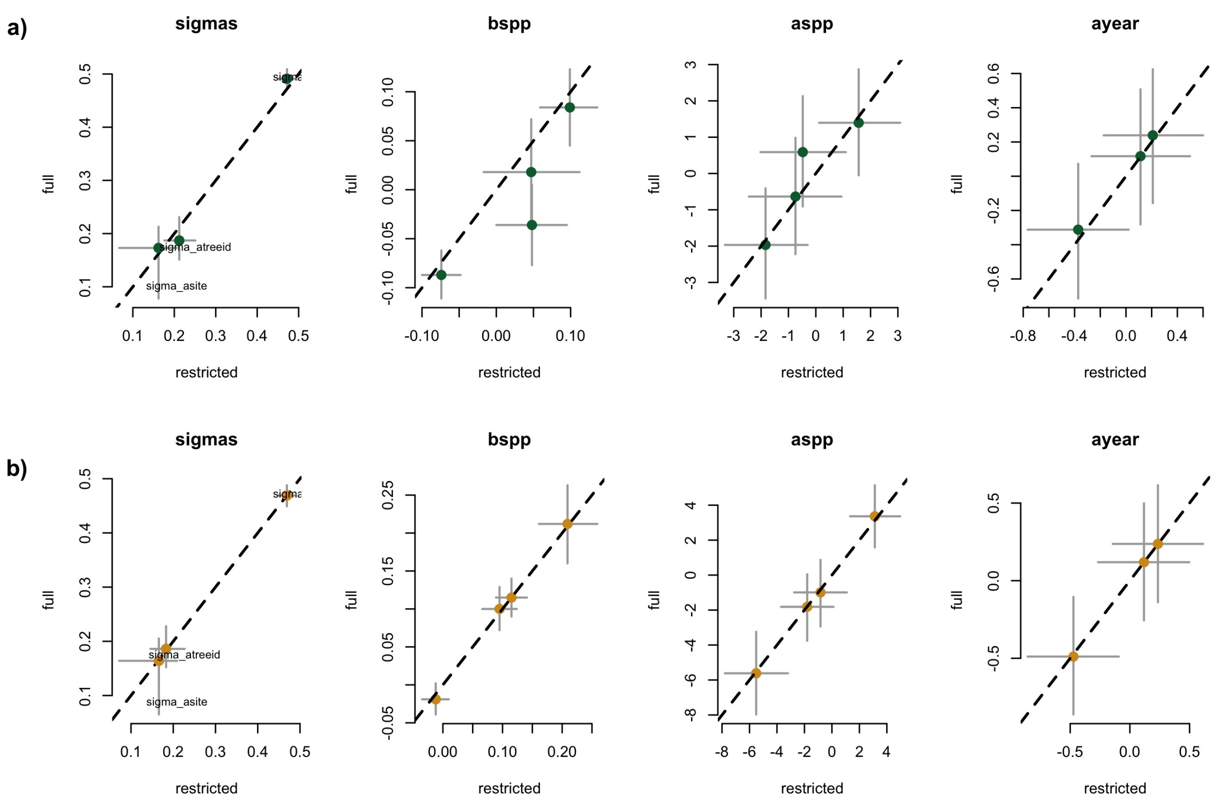

Figure S2: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

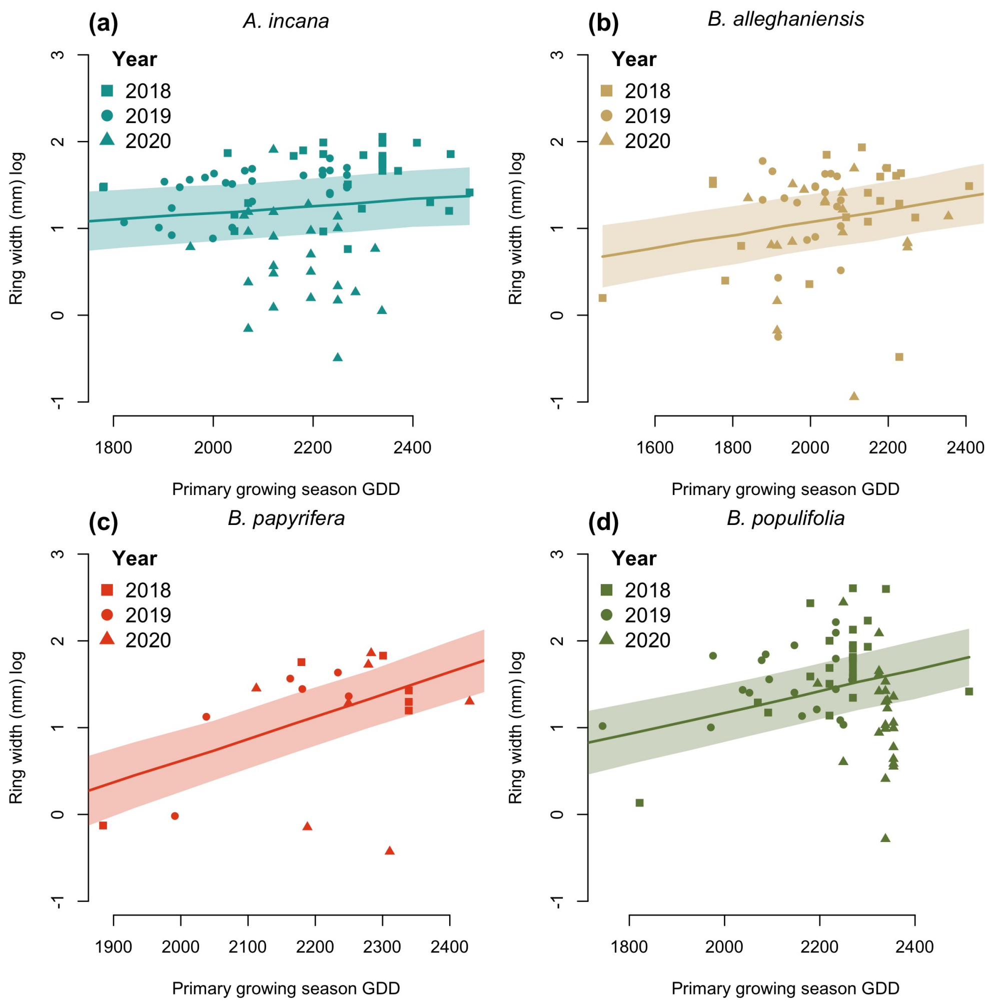

Figure S3: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

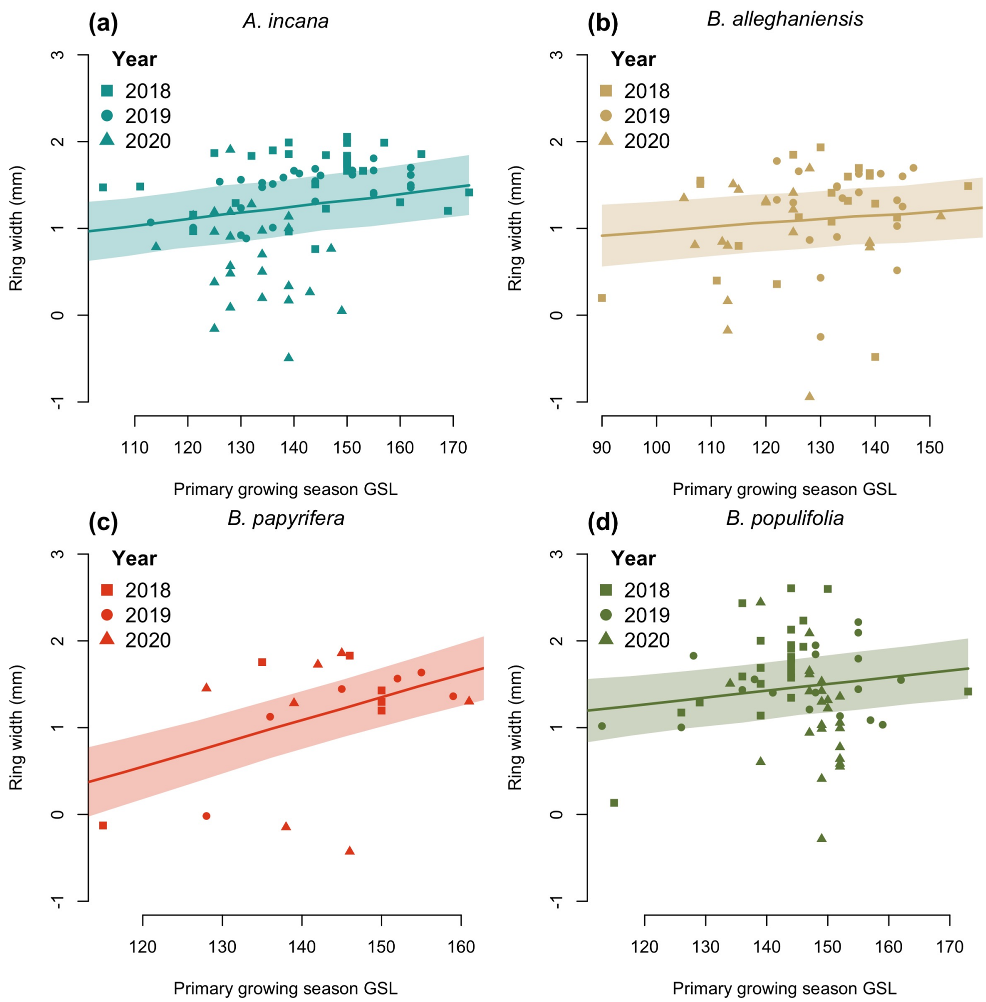

Figure S4: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

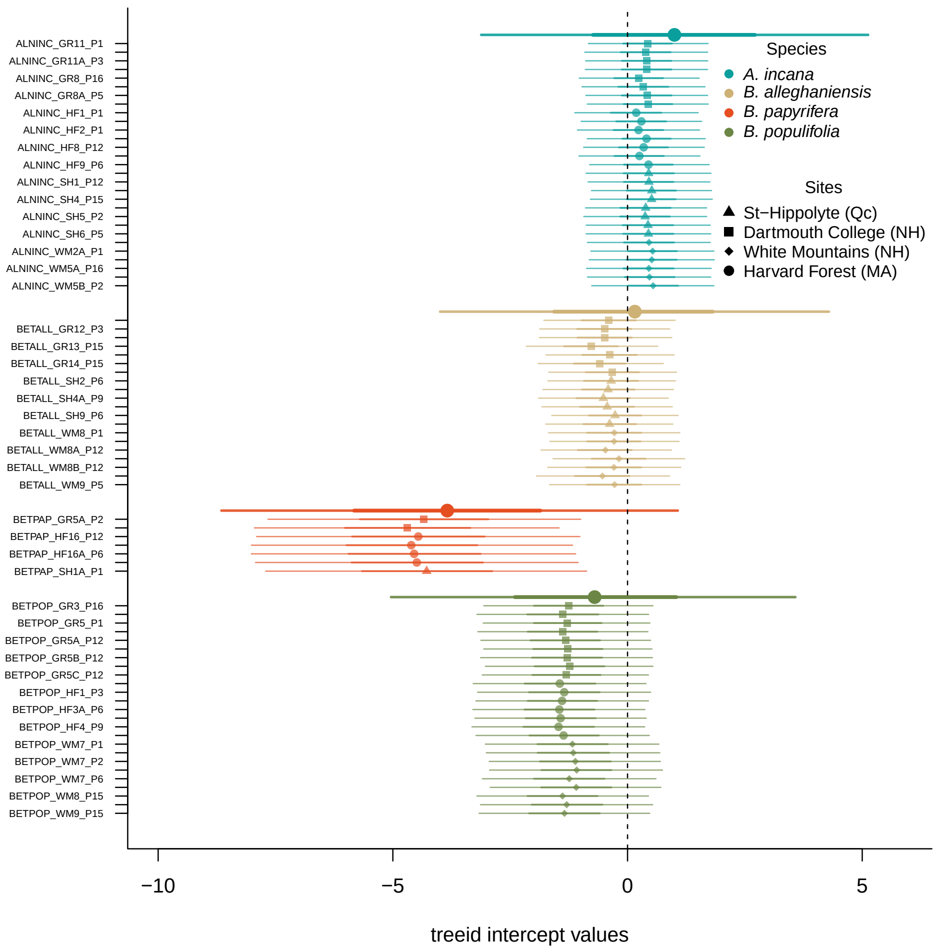

Figure S5: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

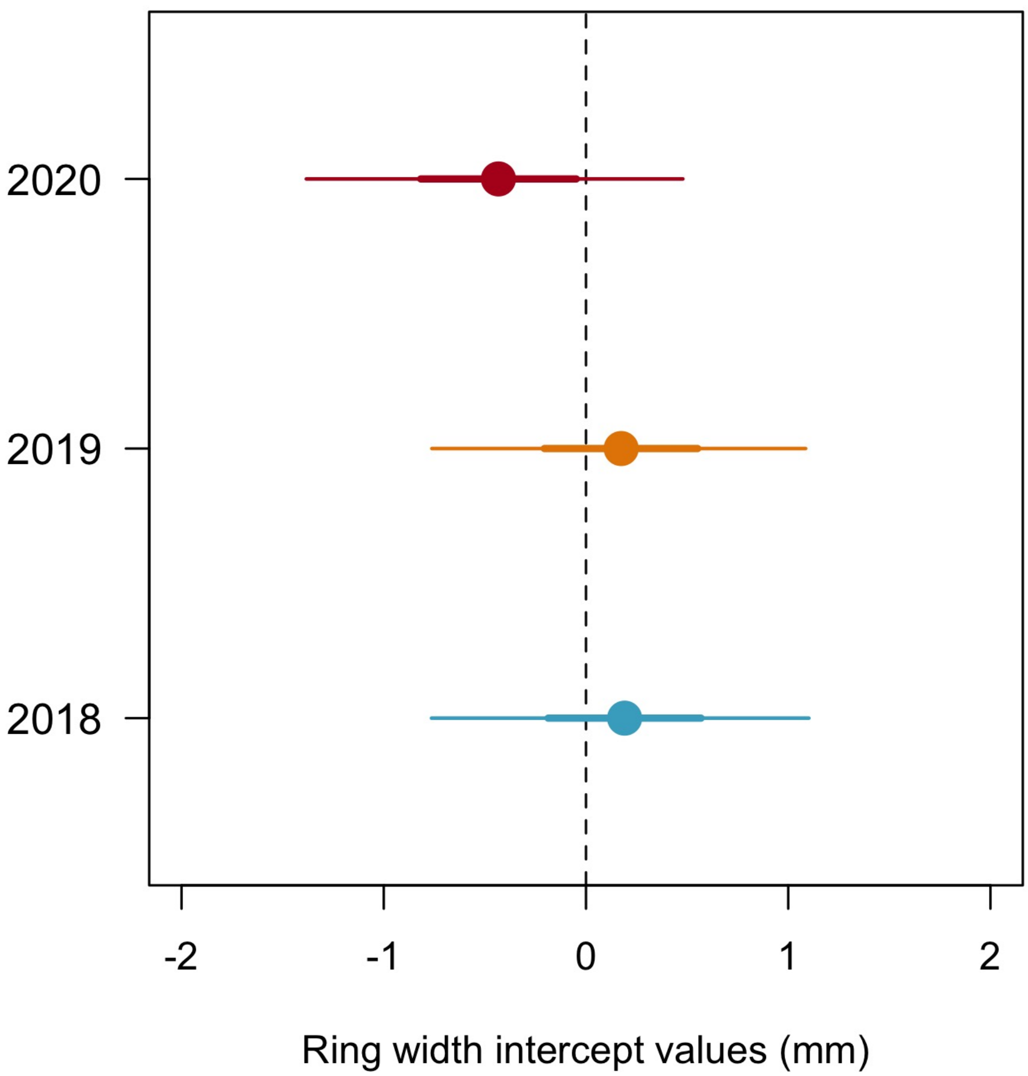

Figure S6: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

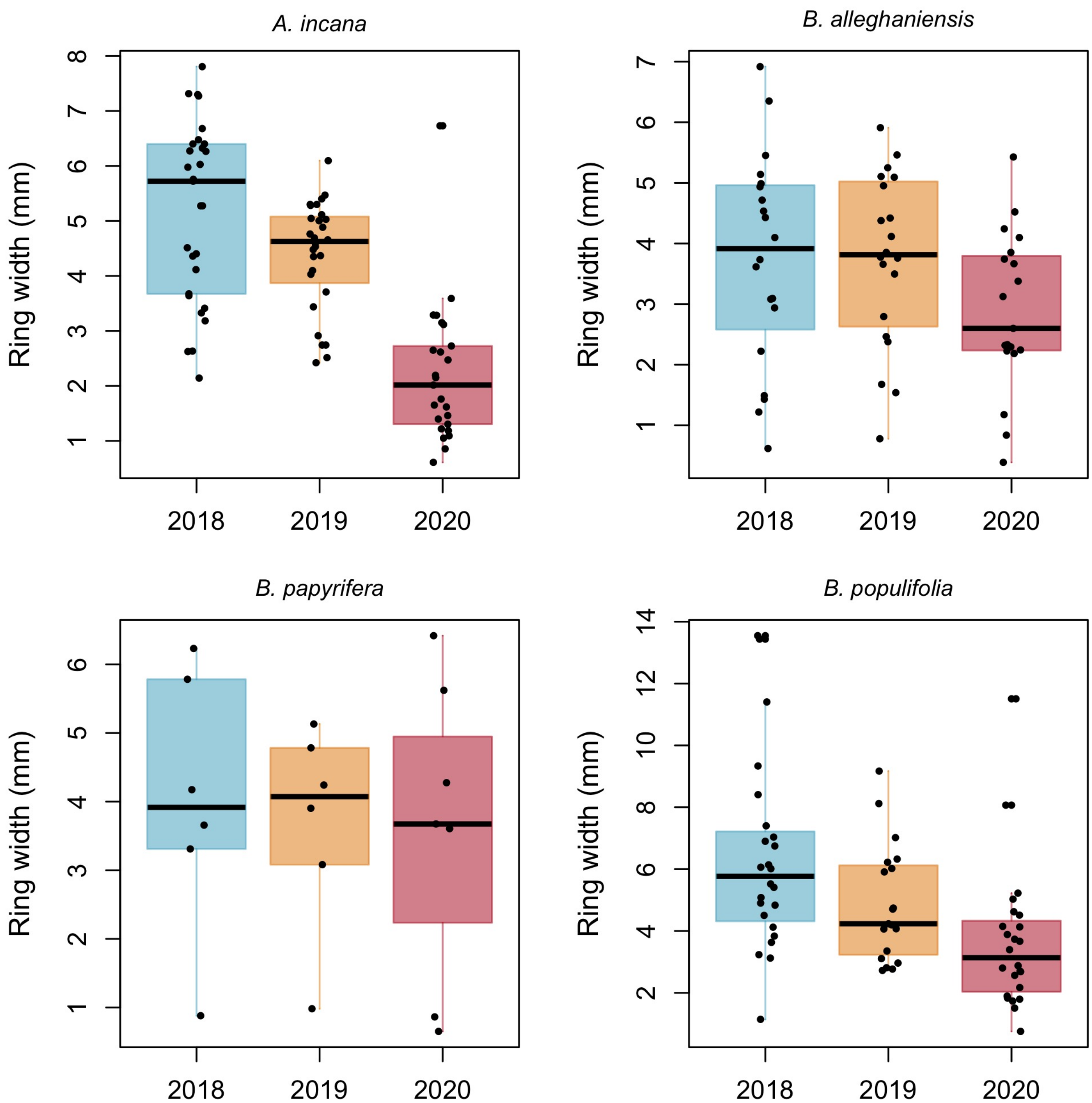

Figure S7: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

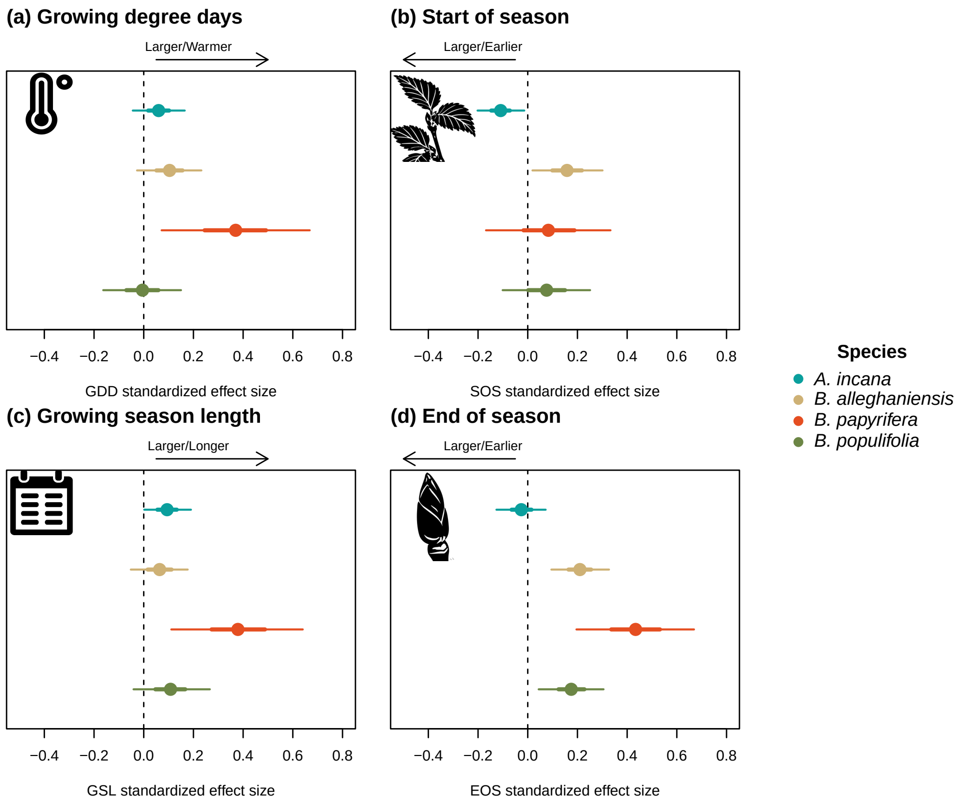

Figure S8: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]